



Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin

Ollscoil Átha Cliath | The University of Dublin

Faculty of Engineering, Mathematics and Science

School of Computer Science & Statistics

Integrated Computer Science
Integrated Engineering
Computer Science and Business
Computer Science and Language

Semester 1 2022

Year 4

CSU44053 Computer Vision

14th December 2022

RDS-SIM COURT

14.00-16.00

Dr Kenneth Dawson-Howe

Instructions to Candidates:

Attempt two questions. All questions carry equal marks. Each question is scored out of a total of 50 marks.

For Application questions, a series of computer vision operations must be detailed to solve the application problem. The input to and output from each technique used must be clearly and precisely stated. How each technique is used within the context of the application must also be described including the setting of any parameters.

For Compare and Contrast type questions marks will only be awarded for the detailed comparison of techniques. No marks will be awarded for separate descriptions of the techniques.

In all questions you must describe computer vision theory and should not refer to code or library calls (OpenCV or any other library).

1. (a) **[APPLICATION QUESTION]** Describe how you would reliably locate emergency exit signs such as those shown the top row of images below. Your solution must consist of a series of computer vision techniques and you must provide details of how the techniques will be applied including expected input and output for each technique. [25 marks]



First row of photos shows some emergency exit signs.



Second row of photos shows some other signs/objects.

1. (b) **[COMPARE & CONTRAST QUESTION]** Compare and contrast:
- Non-Maxima Suppression as used in first derivative edge detection.
 - Non-Maxima Suppression as used in the first stage of SIFT.
 - Non-Maxima Suppression as used in Moravec corner detection.
 - Non-Maxima Suppression as used in the Hough transform for circles where the radius is unknown.

You must provide a list of the differences and similarities between the techniques. Each of the differences and similarities must be clearly explained. **NOTE:** Marks will only be awarded for the detailed comparison of techniques. No marks will be awarded for separate descriptions of the techniques

[25 marks]

2. (a) **[APPLICATION QUESTION]** Using edge detection describe how you would automatically locate a set of downwards steps (such as those shown below). You may assume that the edge of the steps are highlighted by stripes of solid colour which are distinct from the colour of the steps and that the stripes are all the same physical size. You cannot assume any prior knowledge about the colour of the stripes or of the steps or the surrounding area. Your solution must consist of a series of computer vision techniques and you must provide details of how the techniques will be applied including expected input and output for each technique.

[25 marks]



The photos show a series of images of a staircase taken by a camera (frames 10, 43 and 95 from a video sequence) approaching the tops of the stairs. The stripes on each step (yellow & white in this case) will always be a different colour to the colour of the steps, but the colour can vary.

2. (b) **[COMPARE & CONTRAST QUESTION]** Compare and contrast:

- Connected Components Analysis.
- Watershed Segmentation.
- Mean Shift Segmentation.

You must provide a list of the differences and similarities between the techniques. Each of the differences and similarities must be clearly explained. **NOTE:** Marks will only be awarded for the detailed comparison of techniques. No marks will be awarded for separate descriptions of the techniques

[25 marks]

3. (a) **[APPLICATION QUESTION]** Describe how you would locate and recognise bicycle symbols printed on the road such as those shown below. Discuss when your approach might give false positives and what you might do to deal with that. Your solution must consist of a series of computer vision techniques and you must provide details of how the techniques will be applied including expected input and output for each technique.

[25 marks]



First row of photos shows some cycle lane markings on the road



Second row of photos shows some other road markings.

3. (b) **[COMPARE & CONTRAST QUESTION]** Compare and contrast:

- Robust object detection using a cascade of Haar classifiers.
- Scale Invariant Feature Transform.
- Principal Components Analysis.

You must provide a list of the differences and similarities between the techniques. Each of the differences and similarities must be clearly explained. **NOTE:** Marks will only be awarded for the detailed comparison of techniques. No marks will be awarded for separate descriptions of the techniques

[25 marks]